

1、设 A, B 为随机事件, 则 $P(A) = P(B)$ 的充分必要条件是

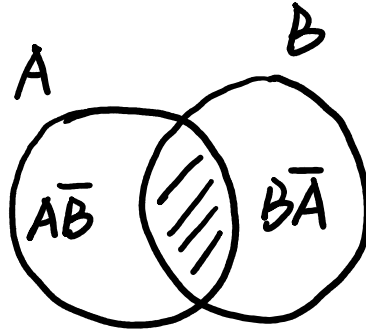
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(A) $P(A \cup B) = P(A) + P(B)$

(B) $P(AB) = P(A)P(B)$

(C) $P(A\bar{B}) = P(B\bar{A})$

(D) $P(AB) = P(\overline{AB})$



(C) $P(A\bar{B}) = P(B\bar{A})$

$\Leftrightarrow P(A\bar{B}) + P(AB) = P(B\bar{A}) + P(AB)$

$\Leftrightarrow P(A) = P(B)$

2、设随机变量 X 与 Y 相互独立, 且都服从正态分布

$N(\mu, \sigma^2)$, 则 $P\{|X - Y| < 1\}$ ()

(A) 与 μ 无关, 与 σ^2 有关 (B) 与 μ 有关, 与 σ^2 无关

(C) 与 μ, σ^2 都有关 (D) 与 μ, σ^2 都无关

$X - Y \sim N(0, 2\sigma^2)$

$P\{|X - Y| < 1\} = P\left\{\left|\frac{X - Y - 0}{\sqrt{2}\sigma}\right| < \frac{1}{\sqrt{2}\sigma}\right\}$

$= \Phi\left(\frac{1}{\sqrt{2}\sigma}\right) - \Phi\left(-\frac{1}{\sqrt{2}\sigma}\right)$

A, B 为二事件, 则 $\overline{A \cup B} = (\quad)$

- (A) AB (B) $\overline{A} \overline{B}$ (C) $A \overline{B}$ (D) $\overline{A} \cup \overline{B}$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$A - B = A \overline{B} = A - AB$$

设 A, B, C 表示三个事件, 则 $\overline{A} \overline{B} \overline{C}$ 表示 ()

- (A) A, B, C 中有一个发生 $A \overline{B} \overline{C} \cup \overline{A} B \overline{C} \cup \overline{A} \overline{B} C$
(B) A, B, C 中恰有两个发生 $A \overline{B} C \cup A \overline{C} B \cup \overline{A} B C$
(C) A, B, C 中不多于一个发生 $A \overline{B} \overline{C} \cup \overline{A} B \overline{C} \cup \overline{A} \overline{B} C$
(D) A, B, C 都不发生

3、 A, B 为两事件, 若 $P(A \cup B) = 0.8, P(A) = 0.2, P(\overline{B}) = 0.4$ 则 () 成立

- (A) $P(A \overline{B}) = 0.32$ (B) $P(\overline{A} \overline{B}) = 0.2$
(C) $P(B - A) = 0.4$ (D) $P(\overline{B} A) = 0.48$

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.8$$

$$0.2 + 0.6 - P(AB) = 0.8 \Rightarrow P(AB) = 0$$

$$(A) P(A \overline{B}) = P(A) - P(AB) = 0.2 - 0 = 0.2$$

$$(B) P(\overline{A} \overline{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - 0.8 = 0.2$$

$$(C) P(B - A) = P(B) - P(AB) = 0.6 - 0 = 0.6$$

$$(D) P(\overline{B} A) = P(\overline{A} \overline{B}) = 0.2$$

6、设离散型随机变量 X 的分布列为

x	0	1	2
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$$P(X=x) = P(X \leq x)$$

6、设离散型随机变量 X 的分布列为

X	0	1	2
p	0.3	0.5	0.2

其分布函数为 $F(x)$, 则 $F(3) = (D)$

- (A)0 (B)0.3 (C)0.8 (D)1

$$F(x) = P(X \leq x)$$

$$F(3) = P(X \leq 3)$$

$$= P(X=0) + P(X=1) + P(X=2)$$

$$= 1$$

连续

7、设离散型随机变量 X 的密度函数为

$$f(x) = \begin{cases} cx^4, & x \in [0,1] \\ 0, & \text{其它} \end{cases}, \text{ 则常数 } c = (D)$$

- (A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C)4 (D)5

$$1 = \int_{-\infty}^{+\infty} f(x) dx = \int_0^1 cx^4 dx = \frac{c}{5} x^5 \Big|_0^1 = \frac{c}{5}$$

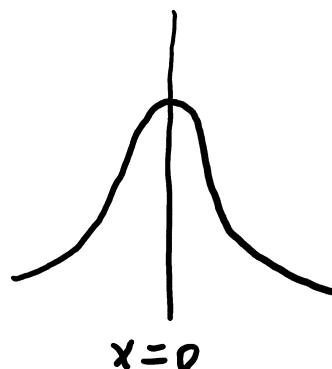
$$\Rightarrow c = 5$$

8、设 $X \sim N(0,1)$, 密度函数 $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, 则

$\varphi(x)$ 的最大值是 (C)

- (A)0 (B)1 (C) $\frac{1}{\sqrt{2\pi}}$ (D) $-\frac{1}{\sqrt{2\pi}}$

$$\varphi(x) = \varphi(0) = \frac{1}{\sqrt{2\pi}}$$



$$\varphi_{\max}(x) = \varphi(0) = \frac{1}{\sqrt{2\pi}}$$

9、设随机变量 X 可取无穷多个值 $0, 1, 2, \dots$, 其概率分布为

$$p(k; 3) = \frac{3^k}{k!} e^{-3}, k = 0, 1, 2, \dots, \text{ 则下式成立的是 ()}$$

$$(A) EX = DX = 3 \quad (B) EX = DX = \frac{1}{3}$$

$$(C) EX = 3, DX = \frac{1}{3} \quad (D) EX = \frac{1}{3}, DX = 9$$

$$X \sim P(3) \Rightarrow EX = 3, DX = 3$$

10、设 X 服从二项分布 $B(n, p)$, 则有 (D)

$$(A) E(2X - 1) = 2np$$

$$(B) D(2X + 1) = 4np(1 - p) + 1$$

$$(C) E(2X + 1) = 4np + 1$$

$$(D) D(2X - 1) = 4np(1 - p)$$

$$X \sim B(n, p) \Rightarrow EX = np, DX = np(1-p)$$

$$(A) E(2X-1) = 2EX-1 = 2np-1$$

$$(B) D(2X+1) = D(2X) = 4DX = 4np(1-p)$$

$$(C) E(2X+1) = 2EX+1 = 2np+1$$

$$(D) D(2X-1) = D(2X) = 4DX = 4np(1-p)$$

11、独立随机变量 X, Y , 若 $X \sim N(1, 4), Y \sim N(3, 16)$,

下式中不成立的是 (CD)

$$(A) E(X + Y) = 4 \quad \checkmark \quad (B) E(XY) = 3 \quad \checkmark$$

$$(C) D(X - Y) = 12 \quad \checkmark \quad (D) E(Y + 2) = 16 \quad \checkmark$$

(A) $E(X+Y) = 4$ ✓ (B) $E(XY) = 3$ ✓

(C) $D(X-Y) = 12$ ✗ (D) $E(Y+2) = 16$ ✗

$$D(X-Y) = DX + DY = 4 + 16 = 20$$

$$E(Y+2) = EY + 2 = 3 + 2 = 5$$

13、设 $X \sim N(0,1)$ ，又常数 c 满足

$$P\{X \geq c\} = P\{X < c\}, \text{ 则 } c = (B)$$

(A) 1 (B) 0 (C) $\frac{1}{2}$ (D) -1

$$\boxed{DX = EX^2 - (EX)^2} \text{ 公式.}$$

14、已知 $EX = -1$, $DX = 3$, 则 $E[3(X^2 - 2)] =$

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(A) 9 (B) 6 (C) 30 (D) 36

$$E(3(X^2 - 2)) = 3E(X^2 - 2) = 3EX^2 - 6$$

$$= 3(DX + (EX)^2) - 6$$

$$= 3(3 + 1) - 6$$

$$= 6$$

17、设 $X \sim b(n, p)$ 且 $EX = 6$, $DX = 3.6$, 则有 (C)

(A) $n = 10, p = 0.6$ (B) $n = 20, p = 0.3$

(C) $n = 15, p = 0.4$ (D) $n = 12, p = 0.5$

17、设 $X \sim b(n, p)$ 且 $EX = 6$, $DX = 3.6$, 则有(C)

(A) $n = 10, p = 0.6$ (B) $n = 20, p = 0.3$

(C) $n = 15, p = 0.4$ (D) $n = 12, p = 0.5$

$$\begin{cases} np = 6 \\ np(1-p) = 3.6 \end{cases} \Rightarrow 1-p = 0.6 \Rightarrow p = 0.4$$

22、设随机变量 $X \sim N(2, 4)$, 且 $\Phi(1.65) = 0.95$.

求 $P(X \geq 5.3)$.

$$\begin{aligned} P(X \geq 5.3) &= P\left(\frac{X-2}{2} \geq \frac{5.3-2}{2}\right) = P\left(\frac{X-2}{2} \geq 1.65\right) \\ &= 1 - P\left(\frac{X-2}{2} \leq 1.65\right) \\ &= 1 - \Phi(1.65) \\ &= 0.05 \end{aligned}$$

23、已知连续型随机变量 ξ 的分布函数为

$$F(x) = \begin{cases} 0, & x \leq 0 \\ \frac{x}{4}, & 0 < x \leq 4 \\ 1, & x > 4 \end{cases}, \text{求 } E\xi, D\xi.$$

它的密度函数 $f(x) = F'(x) = \begin{cases} \frac{1}{4}, & 0 < x < 4 \\ 0, & \text{其他} \end{cases}$

$$E\xi = \int_{-\infty}^{+\infty} x f(x) dx = \int_0^4 x \cdot \frac{1}{4} dx = \frac{1}{8} x^2 \Big|_0^4 = 2$$

$$E\xi^2 = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_0^4 x^2 \cdot \frac{1}{4} dx = \frac{1}{12} x^3 \Big|_0^4 = \frac{16}{3}$$

$$E\xi^2 = \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_0^4 x^2 \cdot \frac{1}{4} dx = \frac{1}{12} x^3 \Big|_0^4 = \frac{16}{3}$$

$$D\xi = E\xi^2 - (E\xi)^2 = \frac{16}{3} - 2^2 = \frac{4}{3}$$

24、设连续型随机变量 X 的分布函数为

$$F(x) = A + B \arctan x, -\infty < x < +\infty.$$

求：(1) 常数 A, B ;

考研

(2) X 落入 $(-1, 1)$ 的概率;

(3) X 的密度函数 $f(X)$.

$$(1) F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = A - B \cdot \frac{\pi}{2} = 0 \quad ①$$

$$F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = A + B \cdot \frac{\pi}{2} = 1 \quad ②$$

由①②解得 $A = \frac{1}{2}$, $B = \frac{1}{\pi}$, $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$

$$(2) P(-1 < X < 1) = F(1) - F(-1) = \left(\frac{1}{2} + \frac{1}{\pi} \cdot \frac{\pi}{4}\right) - \left(\frac{1}{2} - \frac{1}{\pi} \cdot \frac{\pi}{4}\right) \\ = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$(3) f(x) = F'(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < +\infty$$

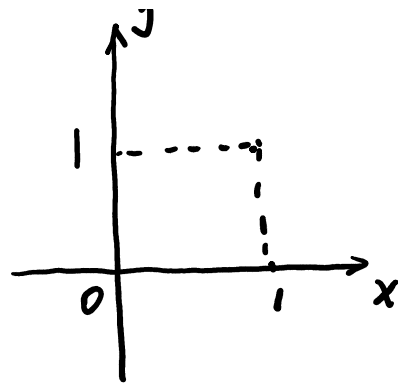
26、设 (ξ, η) 的联合密度为

$$f(x, y) = 4xy, \quad 0 < x, y < 1$$

↑
y

26、设 (ξ, η) 的联合密度为

$$p(x, y) = \begin{cases} 4xy, & 0 \leq x, y \leq 1 \\ 0, & \text{其它} \end{cases}$$



求: (1) 边际密度函数 $p_\xi(x)$, $p_\eta(y)$;

(2) $E\xi$, $E\eta$;

(3) ξ 与 η 是否独立.

(1) $x < 0$ 或 $x > 1$ 时, $p_\xi(x) = 0$

$$0 \leq x \leq 1 \text{ 时, } p_\xi(x) = \int_{-\infty}^{+\infty} p(x, y) dy = \int_0^1 4xy dy = 2x$$

$$\text{故 } p_\xi(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{其他} \end{cases} \quad \text{同理, } p_\eta(y) = \begin{cases} 2y, & 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$(2) E\xi = \int_{-\infty}^{+\infty} x p_\xi(x) dx = \int_0^1 x \cdot 2x dx = \frac{2}{3} \quad \text{同理, } E\eta = \frac{2}{3}$$

(3) $p(x, y) = p_\xi(x) \cdot p_\eta(y)$ 故 x 与 y 独立.

27、设 X_1, X_2 是来自正态总体 $N(\mu, 1)$ 的样本, 下列三个估

计量是不是参数 μ 的无偏估计量, 若是无偏估计量, 试判断

哪一个较优?

$$\hat{\mu}_1 = \frac{2}{3}X_1 + \frac{1}{3}X_2, \quad \hat{\mu}_2 = \frac{1}{4}X_1 + \frac{3}{4}X_2,$$

$$\hat{\mu}_3 = \frac{1}{2}X_1 + \frac{1}{2}X_2.$$

$$E\hat{\mu}_1 = E\left(\frac{2}{3}X_1 + \frac{1}{3}X_2\right) = \frac{2}{3}EX_1 + \frac{1}{3}EX_2 = \frac{2}{3}\mu + \frac{1}{3}\mu = \mu$$

$$E\hat{\mu}_2 = E\left(\frac{1}{4}X_1 + \frac{3}{4}X_2\right) = \frac{1}{4}EX_1 + \frac{3}{4}EX_2 = \frac{1}{4}\mu + \frac{3}{4}\mu = \mu$$

$$E\hat{\mu}_3 = E\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{2}EX_1 + \frac{1}{2}EX_2 = \frac{1}{2}\mu + \frac{1}{2}\mu = \mu$$

... $\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_3$ 都是 μ 的无偏估计量

故 $\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_3$ 均为 μ 的无偏估计量.

$$D\hat{\mu}_1 = D\left(\frac{2}{3}X_1 + \frac{1}{3}X_2\right) = \frac{4}{9}DX_1 + \frac{1}{9}DX_2 = \frac{5}{9}$$

$$D\hat{\mu}_2 = D\left(\frac{1}{4}X_1 + \frac{3}{4}X_2\right) = \frac{1}{16}DX_1 + \frac{9}{16}DX_2 = \frac{5}{8}$$

$$D\hat{\mu}_3 = D\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{4}DX_1 + \frac{1}{4}DX_2 = \frac{1}{2}$$

由于 $D\hat{\mu}_3 < D\hat{\mu}_1$ 且 $D\hat{\mu}_3 < D\hat{\mu}_2$. 故 $\hat{\mu}_3$ 最优.

28、设 $\xi \sim f(x, \theta) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}}, & x > 0 \ (\theta > 0), \\ 0, & \text{其它} \end{cases}$

$x_1, x_2, \dots, x_n$ 为 ξ 的一组观察值, 求 θ 的极大似然估计

$$\text{似然函数 } L(\theta) = \prod_{i=1}^n \frac{1}{\theta} e^{-\frac{x_i}{\theta}} = \frac{1}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i}$$

$$\text{取对数 } \ln L(\theta) = -n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n x_i$$

$$\text{令 } \frac{d \ln L(\theta)}{d\theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n x_i = 0 \text{ 得 } \hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

设 $D(X) = 16$, $D(Y) = 25$, $\rho_{XY} = 0.3$, 则

$$D(X + Y) = \underline{\hspace{2cm}}.$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{DX} \cdot \sqrt{DY}} \Rightarrow \text{Cov}(X, Y) = 0.3 \times \sqrt{16} \times \sqrt{25} = 6$$

$$\begin{aligned} D(X+Y) &= DX + DY + 2\text{Cov}(X, Y) \\ &= 16 + 25 + 12 \\ &= 53 \end{aligned}$$

1、设 A, B 为随机事件,

$$P(A) = 0.5, P(B) = 0.6, P(B|A) = 0.8,$$

$$\text{则 } P(B \cup A) = \underline{\hspace{2cm}}.$$

$$P(AB) = P(A)P(B|A) = 0.4$$

$$\begin{aligned} P(B \cup A) &= P(B) + P(A) - P(AB) \\ &= 0.6 + 0.5 - 0.4 \\ &= 0.7 \end{aligned}$$

8、设 $X_1, X_2, \dots, X_5$ 是来自总体 $X \sim N(0, 1)$ 的简单随机样

本, 统计量 $C(X_1 + X_2) / \sqrt{X_3^2 + X_4^2 + X_5^2} \sim t(n)$, 则常数 $C = \underline{\frac{\sqrt{6}}{2}}$, 自由度 $n = \underline{3}$.

$$X_1 + X_2 \sim N(0, 2) \Rightarrow \frac{X_1 + X_2}{\sqrt{2}} \sim N(0, 1)$$

$$\underline{X_1 + X_2}$$

$$\sqrt{2}$$

$$X_1 + X_2$$

$$\frac{\frac{X_1 + X_2}{\sqrt{2}}}{\sqrt{\frac{X_3^2 + X_4^2 + X_5^2}{3}}} = \underbrace{\frac{\sqrt{3}}{\sqrt{2}}}_{C = \frac{\sqrt{6}}{2}} \cdot \frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}} \sim t(3)$$

10、设顾客在某银行窗口等待服务的时间（以分计） X 服从指数分布，其概率密度函数为

$$f(x) = \begin{cases} \frac{1}{5} e^{-x/5} & x > 0 \\ 0, & \text{其它} \end{cases}$$

考研竞赛数学

某顾客在窗口等待服务，若超过 10 分钟，他就离开。他一个月到银行 5 次。以 Y 表示一个月内他未等到服务而离开窗口的次数，写出 Y 的分布律，并求 $P\{Y \geq 1\}$ 。

$$\begin{aligned} P(\text{顾客未等到服务离开}) &= P(X > 10) = 1 - P(X \leq 10) \\ &= 1 - \int_0^{10} \frac{1}{5} e^{-x/5} dx = 1 + \int_0^{10} e^{-x/5} d(-\frac{x}{5}) \\ &= 1 + e^{-x/5} \Big|_0^{10} = e^{-2} \end{aligned}$$

Y 的可能取值为 0, 1, 2, 3, 4, 5

$$P(Y=0) = (1 - e^{-2})^5 \quad P(Y=1) = C_5^1 e^{-2} \cdot (1 - e^{-2})^4$$

...

$$P(Y \geq 1) = 1 - P(Y=0) = 1 - (1 - e^{-2})^5$$